



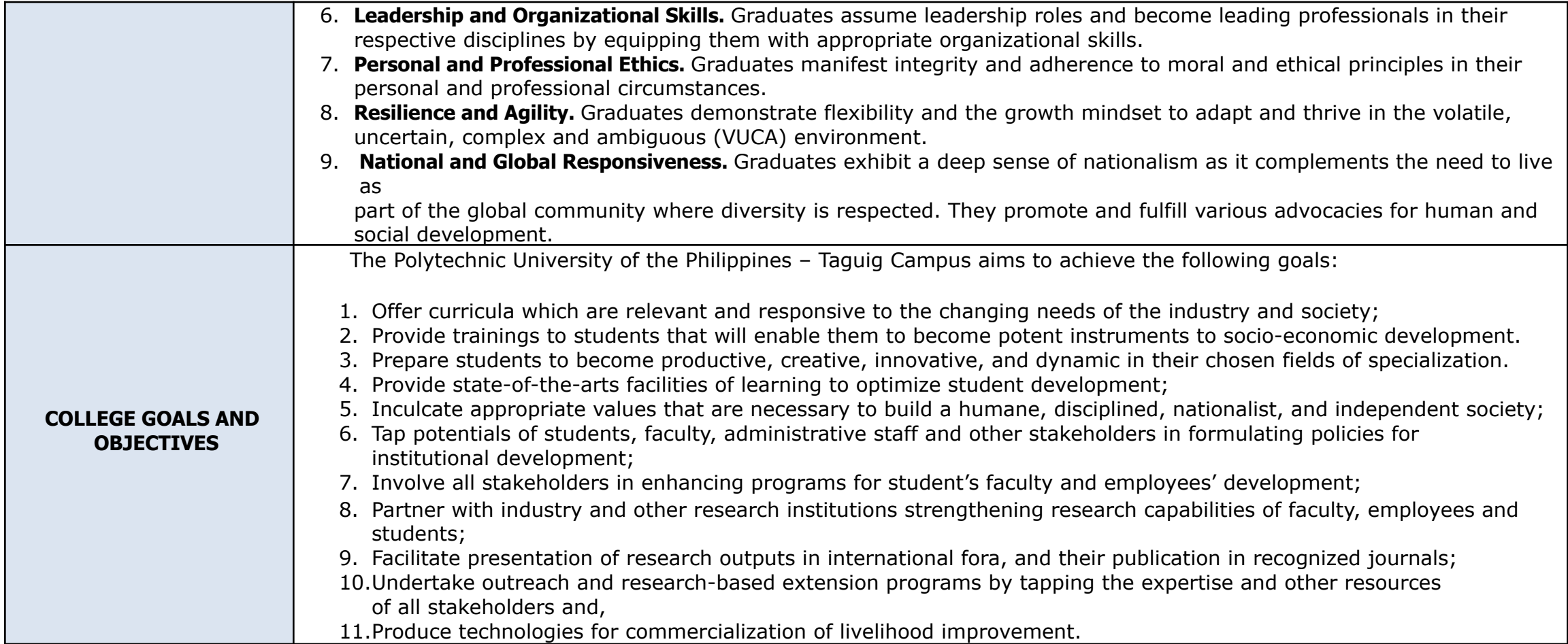
COURSE SYLLABUS						
Course Code	ITEC 301	Course Title	Advance Programming		Course Credit	3
Course Description		The course aims to teach a high level programming language that is beneficial to a variety of real world applications. It introduces the advanced topics of the selected language as well as its properties. It will also involve development of several applications, case studies and exercises. During the second half of the semester, students work in teams to build a mid-scale software application based on the needs of a specific client. Each team presents the final project, sharing how course concepts were applied to address unique client needs. The concepts and skills learned in this course are transferable to a wide variety of contemporary programming languages and software development tools.				
Pre-Requisites		COMP 003	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL LEARNING
OUTCOMES
(ILO)**

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.

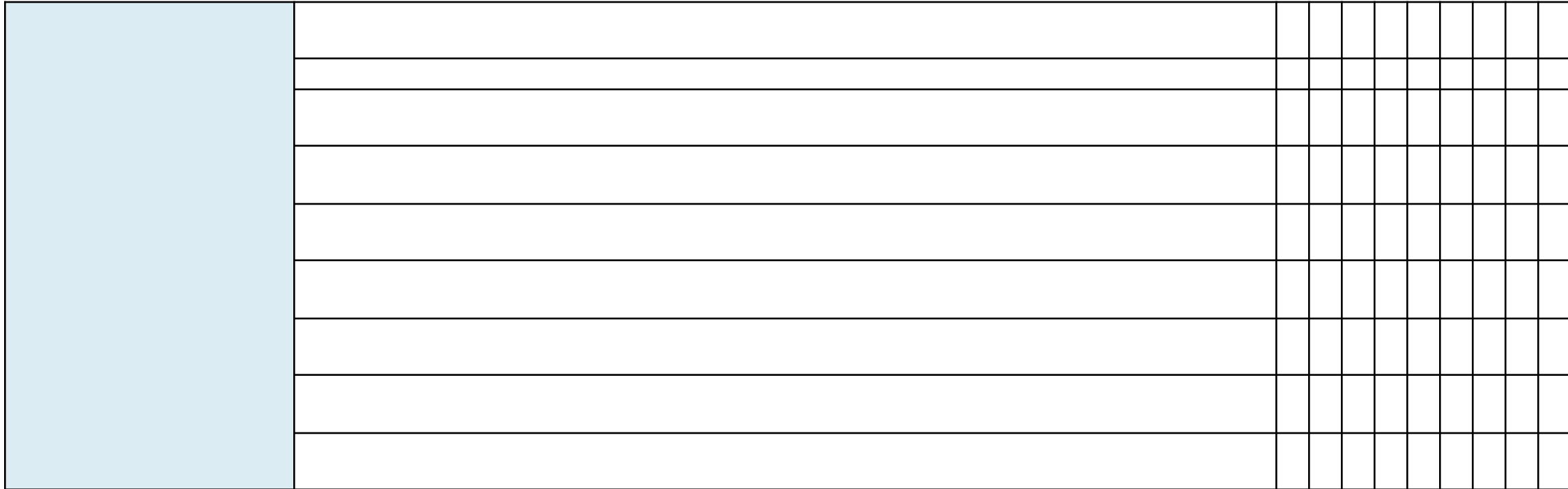




PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO							Alignment to ILOs								
								1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.															
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution															
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.															
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.															
		Integrate IT-based solutions into the user environment effectively.														
		Assist in the creation of an effective IT project plan.														
		Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession														
		Understand best practices and standards and their applications.														
		Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.														



		Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.																	
		Analyze the local and global impact of computing information technology on individuals, organizations, and society.																	
		Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.																	
		Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development																	



COURSE LEARNING OUTCOMES (CLO)	Based on CMO	Alignment to PLOs							
		1	2	3	4	5	6	7	8

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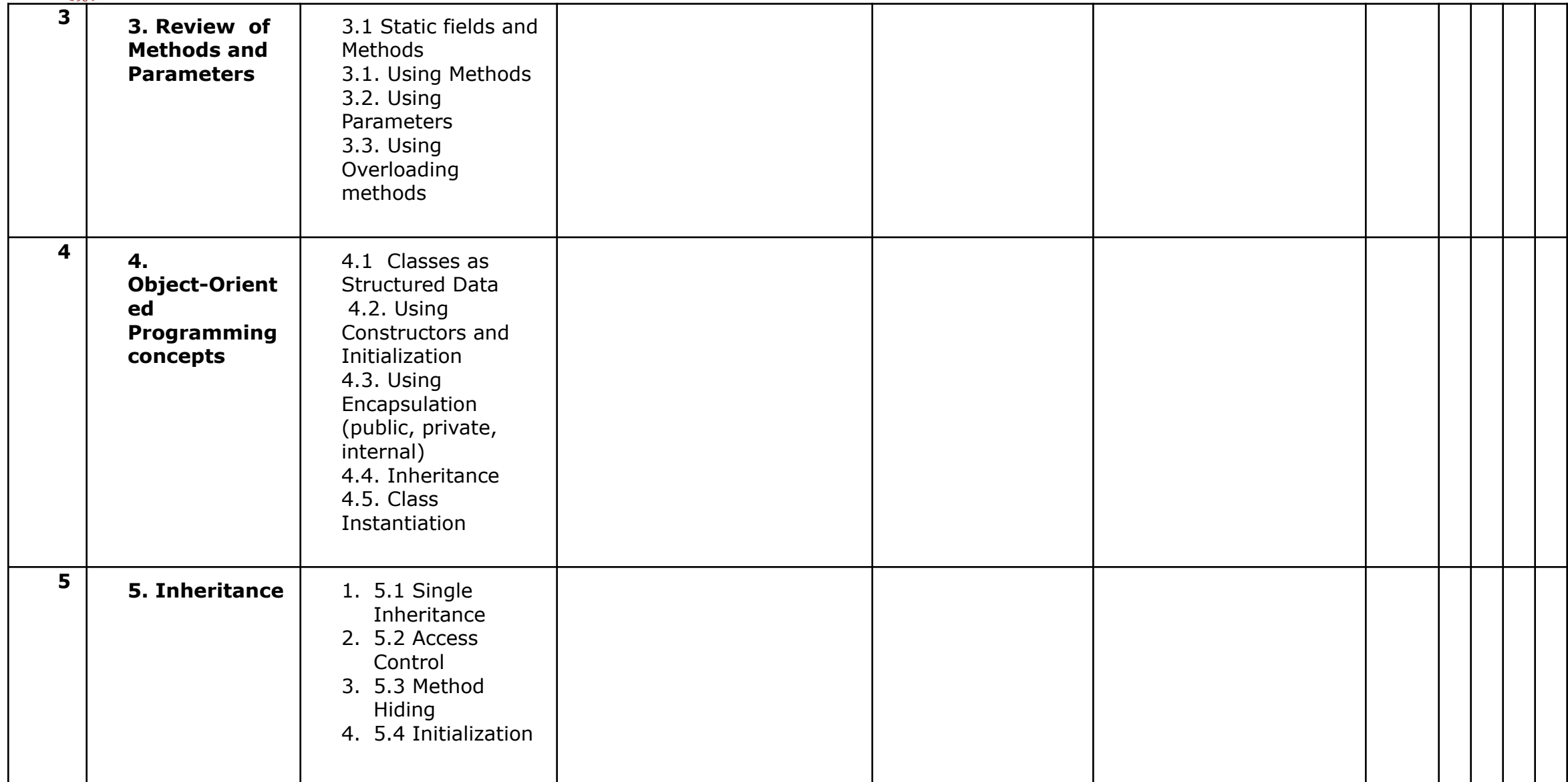
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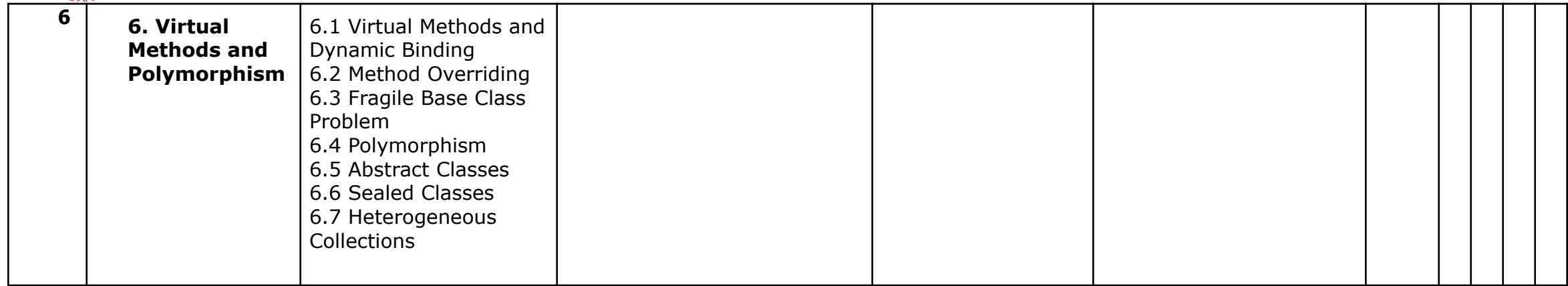
OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignmen t to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLT As	1	2	3	4
1	1. Review of Statements and Exceptions	1.1 Using Selection Statements (if, else, switch) 1.2 Using Iteration Statements (for, while, do 1.3 While, foreach 1.4 Using Jump Statements (break, continue, goto) 1.5 Handling Basic Exceptions					X			
2	2. Review of Arrays and Introduction of Indexers	2.1. Overview of Single and Rectangular 2.2 Creating Jagged arrays 2.3 Indexers								



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MIDTERM WRITTEN EXAMINATION									
7	7. Interfaces	1. Interface Fundamentals 2. Programming with Interfaces 3. Using Interfaces at Runtime							
8	8 . Advanced Windows Application	8.1 Menus, Tabbed windows and Multiple document interface (MDI) programs. 8.2 ListView and TreeView controls 8.3 Display lists of information in ListBox, CheckedListBox and							

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		ComboBox controls. 8.4 Create custom controls.							
9	9. Data Connection	9.1 ADO.NET with RDBMS and SQL - Single Table - Multiple Tables							
10	10. File Handling	10.1 File Classes and Directory 10.2 Sequential-access file processing. 10.3 FileStream classes, StreamReader and StreamWriter 10.4. FileStream classes and BinaryFormatter							
	FINAL EXAMINATION								
18	FINAL EXAMINATION (Written Exam)								



POLYTECHNIC UNIVERSITY OF THE
PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR
CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



LIST OF REFERENCES

BOOKS:

WEBSITE:

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General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

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CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

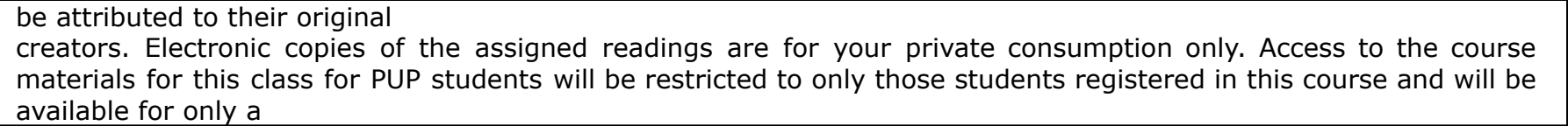
Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will

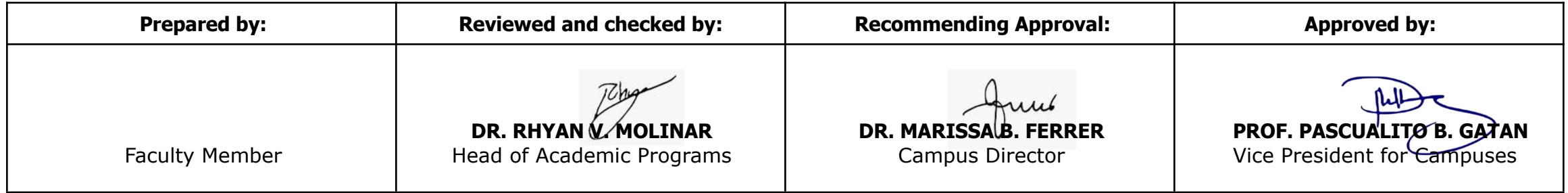


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	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.		
COURSE REQUIREMENT/S	GRADING SYSTEM		
Students are required to attend classes regularly, complete all programming exercises and assessments, and develop a final COBOL project demonstrating structured programming concepts. Active participation, timely submission of tasks, and adherence to academic integrity are expected throughout the course.	Mid Term Class Standing 70% Quizzes Class Recitation Seatworks Case Studies Assignment Midterm Examinations 100%	Final Term Class Standing 70% Quizzes Class Recitation Seatworks Case Studies Assignment Final Examinations 30% 100%	
	Final Grade = (Midterm Grade + Final Term Grade) / 2		
CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section:		Name of Faculty:	
Day and Time:		Consultation Time:	
Room:		Office Tel. No./ Mobile Phone No.	
School Year:	Semester:	E-Mail Address:	



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